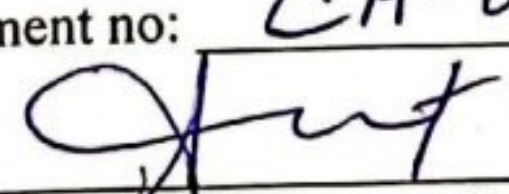


	NON-CONFORMITY REPORT FOR ASSESSMENT VISIT	F-02/06
		Issue Date: 11/11/19 Rev No: 06

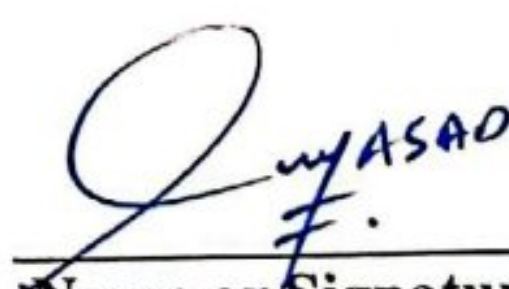
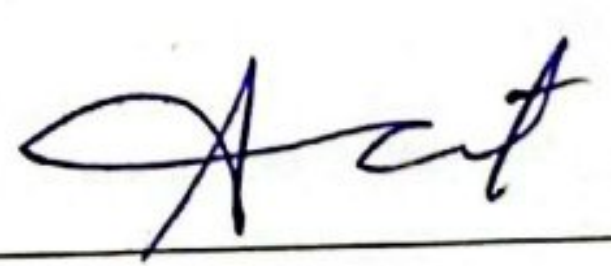
ACTIVITY:	☒ Assessment	☐ Surveillance	NCR No: 6
	☐ Re-Assessment	☐ Scope extension	Accr./Appl.No:
	☐ Other Please specify:		Non-conformity category: Very serious ☐ Essential ☒
			Time limit for correction: 6 weeks
ORGANIZATION:	Quality control Lab vital Agri. Nutrients (Pvt) Ltd.		Mr. Hammyat Hussain
Department:	Labs		
Team Leader.	Mr. Atique Ur Rehman	Mngt. Rep.	
Assessor	Dr. Muhammad Aftab		
Expert			
DESCRIPTION: The lab has not developed a procedure for validation and verification of methods (NC). The raw data regarding the method validation is not maintained (NC). 10-01-2025 Date  Signature Assessor/Expert  Signature (Management Representative)			Requirement of: ISO/IEC 17025 ☒ ISO 15189 ☐ ISO/IEC 17021 ☐ ISO/IEC 17020 ☐ Others:
			Ref. Organization's doc.
CORRECTIVE ACTIONS: (Note: It is mandatory to attach Root Cause Analysis) Corrective action for shortcomings on both the NCs has been taken by developing the procedure for Validation and Verification of methods and maintained the raw data regarding method validation.			☐ It is not necessary to attach documentation
Action are documented in the amendment no: <u>CA-06-02-2025</u> <u>14-02-25</u> Date  Signature (Management Representative)			
REASON FOR CLOSING: (To be filled in by the Team Leader.) ☐ The non-conformity is closed based on satisfactory documentation from the organization ☐ The non-conformity is closed based on recommendation from the technical assessor ☐ Implementation of the corrective actions will be followed up at the next visit			
Remarks (if any) ----- 			
The non-conformity is closed: _____ Date Signature (Team Leader.)			

	CORRECTIVE ACTION REQUEST	Doc. No.	QCL-FRM-9.01
		Issue Date	01-10-2024
		Rev#	01

Initiated by: Masab Rasheed	Department: QC Lab	CAR No. CA-06-02-2025
Designation: AQCM	CAR Initiated Date: 07-02-2025	NC No. 06

CAR Initiated due to (Please tick the relevant appropriate box below)

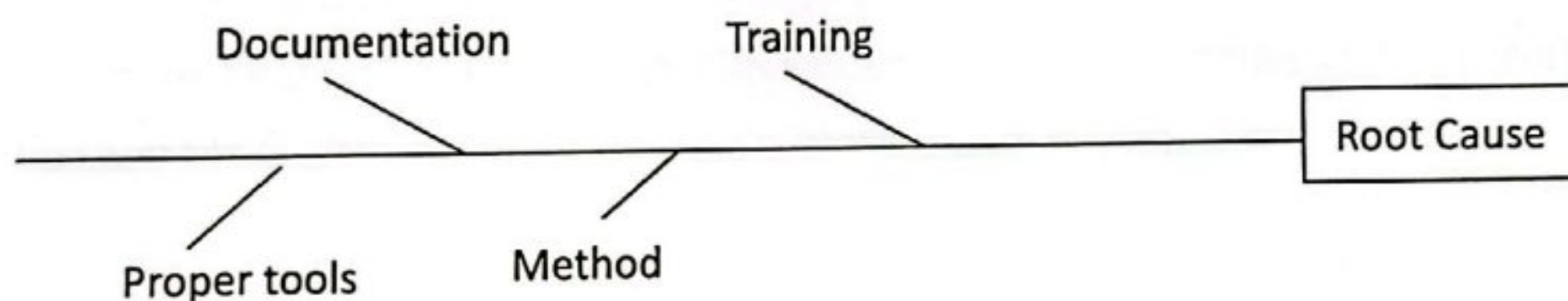
- ☒ PNAC Assessment
 ☐ Complaint
 ☐ Accident/Incident
 ☐ Internal Audit
☐ Suggestion/Improvement
 ☐ Technical Fault
 ☐ Others

A. Description (to be filled by Initiator) The lab has not developed a procedure for validation and verification of methods (NC). The raw data regarding the method validation is not maintained (NC).  Signatures		
Significance/Priority Level	☒ Urgent ☐ Normal ☐ Non-significance	
☒ Accepted ☐ Rejected	Assigned to: Masab Rasheed Date: 07-02-2025	
* If accepted then Manager QC to mark it to concerned person for Root Cause Analysis.		
B. Root Cause Analysis (where required) 1. The lab management failed to establish clear responsibilities and oversight for the development of the procedure due to insufficient awareness and training on the importance of method validation and verification as part of the quality management system. 2. Insufficient training and a lack of clear procedures for maintaining raw data due to inadequate management focus on compliance requirements.		
C. Proposed Action	Target Date:	14-02-2025
Corrective action for shortcomings on both the NCs has been taken by developing the procedure for Validation and Verification of methods and maintained the raw data regarding method validation.  Name or Signatures (Person who carried out Root Cause Analysis and Proposed Corrective Action)		
D. Verification by Manager QC for implementation & Effectiveness		
Is Proposed Action Implemented? ☒ YES ☐ NO	Is it Effective? ☒ YES ☐ NO	
Remarks (If any): CAR closing date: <u>14-02-25</u> Manager QC Sign: 		

	ROOT CAUSE ANALYSIS FORM	Doc. No.	QCL-FRM-9.04
		Issue Date	01-10-2024
		Rev#	01

Non-Conformance Description			
The lab has not developed a procedure for validation and verification of methods (NC).			
RCA Team Members	Himayat Hussain (QCM), Masab Rasheed (AQCM)		
RCA Team Lead / Head	Himayat Hussain (QCM)		
RCA Start Date	07-02-2025	Submission Date	14-02-2025
Methodology / Tools Used			
Cause & Effect Diagram	☐	5WHY analysis	☐
		3W2H / 5W2H	☒

Cause Effect Diagram

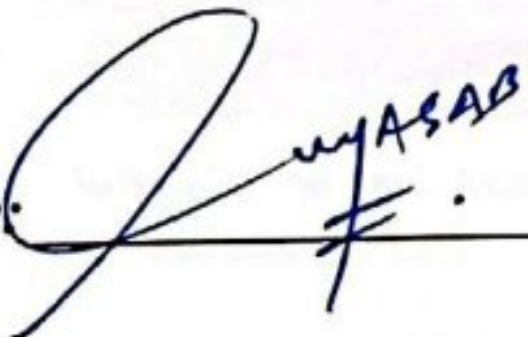


3W2H / 5W2H	Phenomenon of Problem / Question	Explanation / Answer
WHY	Why was there no procedure for validation and verification of methods?	The lab management did not prioritize creating this procedure.
WHY	Why was the procedure not prioritized?	There was a lack of awareness or training on the importance of validation and verification processes.
WHY	Why was there a lack of awareness or training?	The management team failed to include method validation and verification in their quality management training program.
WHY	Why did the management fail to include this in the training program?	No specific individual or team was assigned responsibility for ensuring the completeness of the training program.
WHY	Why was responsibility not assigned?	The lab's organizational structure did not include clear accountability for quality system development and compliance.
How	How can this be prevented from recurring?	Assign a qualified person/team (e.g., Quality Control Officer) to oversee the development of standard

	ROOT CAUSE ANALYSIS FORM	Doc. No.	QCL-FRM-9.04
		Issue Date	01-10-2024
		Rev#	01

		operating procedures (SOPs) for method validation and verification. • Include method validation and verification topics in regular training sessions for lab personnel.
How	How will the effectiveness of corrective actions be ensured?	• Conduct internal audits to ensure the procedure for validation and verification of methods is developed, implemented, and followed. • Regularly review training records and SOPs during management review meetings to confirm compliance and effectiveness.
Root Cause: The lab management failed to establish clear responsibilities and oversight for the development of the procedure due to insufficient awareness and training on the importance of method validation and verification as part of the quality management system.		

RCA Statement: The lab management failed to establish clear responsibilities and oversight for the development of the procedure due to insufficient awareness and training on the importance of method validation and verification as part of the quality management system.

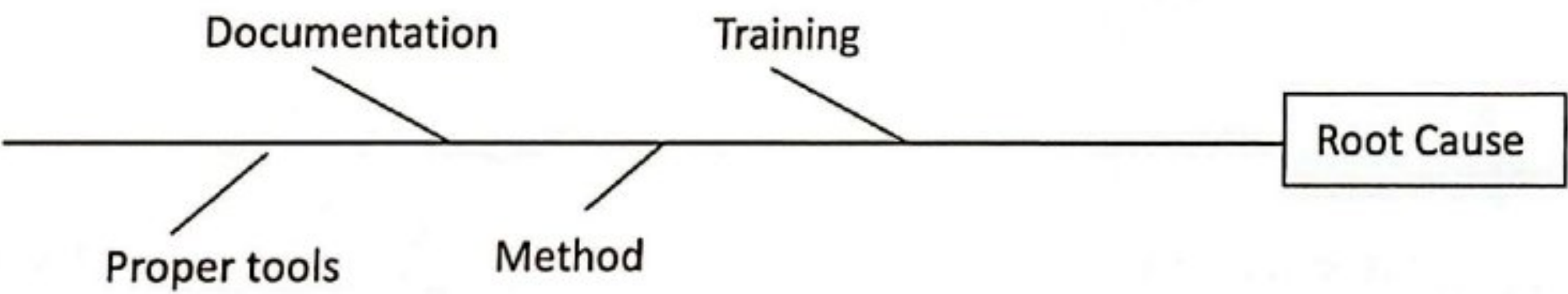
Reviewed by: 

Evaluated by: 

	ROOT CAUSE ANALYSIS FORM	Doc. No.	QCL-FRM-9.04
		Issue Date	01-10-2024
		Rev#	01

Non-Conformance Description			
The raw data regarding the method validation is not maintained (NC).			
RCA Team Members	Himayat Hussain (QCM), Masab Rasheed (AQCM)		
RCA Team Lead / Head	Himayat Hussain (QCM)		
RCA Start Date	07-02-2025	Submission Date	14-02-2025
Methodology / Tools Used			
Cause & Effect Diagram	☐	5WHY analysis	☐
		3W2H / 5W2H	☒

Cause Effect Diagram

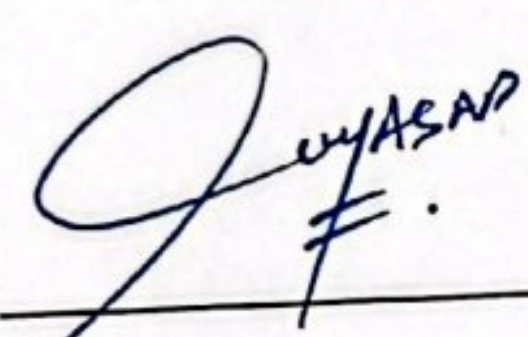


3W2H /5W2H	Phenomenon of Problem / Question	Explanation / Answer
WHY	Why did the raw data regarding the method validation not get maintained?	There was no clear procedure or guideline for recording and maintaining the data.
WHY	Why was there no clear procedure or guideline for data maintenance?	The team did not consider data maintenance critical for the process.
WHY	Why was data maintenance not considered critical for the process?	There was a lack of training and awareness regarding regulatory and quality requirements.
WHY	Why was there a lack of training and awareness regarding regulatory requirements?	The training program for the team was insufficient and not properly reviewed for gaps.
WHY	Why was the training program insufficient?	Management did not prioritize continuous improvement and resource allocation for training.
How	How can we ensure the raw data regarding method validation is maintained going forward?	- Develop and implement a detailed Standard Operating Procedure (SOP) for recording and maintaining raw data. - Provide regular training sessions for employees on the importance of raw

	ROOT CAUSE ANALYSIS FORM	Doc. No.	QCL-FRM-9.04
		Issue Date	01-10-2024
		Rev#	01

		data maintenance and regulatory compliance.
How	How can we prevent this issue from recurring?	• Conduct periodic internal audits to ensure compliance with the new SOP. • Incorporate data maintenance requirements into routine performance checks and quality metrics. • Allocate sufficient resources for training and documentation tools.
Root Cause: Insufficient training and a lack of clear procedures for maintaining raw data due to inadequate management focus on compliance requirements.		

RCA Statement: Insufficient training and a lack of clear procedures for maintaining raw data due to inadequate management focus on compliance requirements.

Reviewed by: 

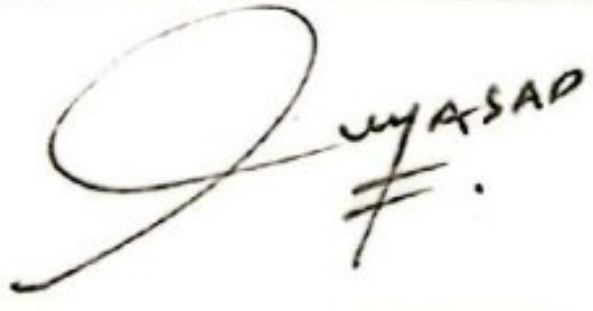
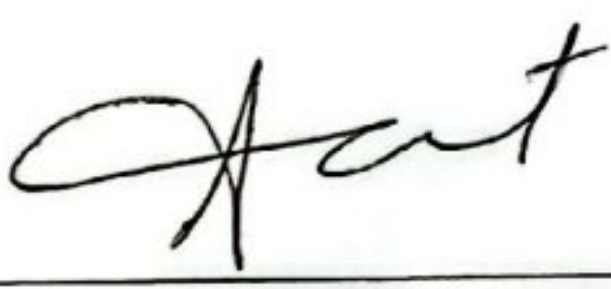
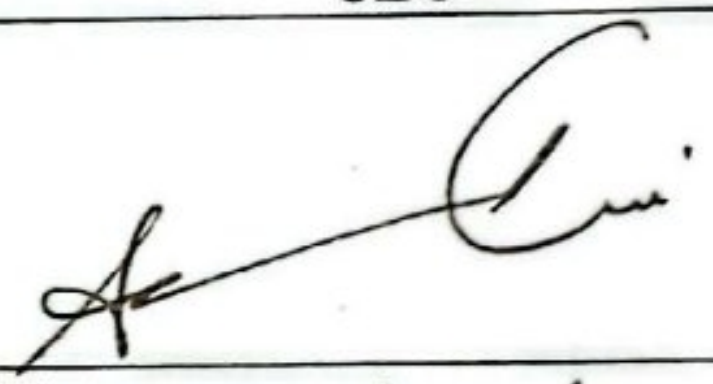
Evaluated by: 

 MASTER	TEST METHODS AND METHOD VALIDATION	Doc. No.	QCL-LSP-23
		Issue Date	10-02-2025
		Rev#	00

CONTROLLED COPY
NO. 1

**Standard Operating Procedure for
Test Methods and Method Validation**

ISO/IEC 17025:2017 – Clause 7.2

Prepared By: Masab Rasheed	Reviewed By: Himayat Hussain	Approved By: Ahmad Umair
AQCM	QCM	CEO
		
Date: 10-02-25	Date: 10-02-25	Date: 10/02/2025

Copyright

The copyright of this procedure, together with all confidential information contained herein is the sole property of **VAN QC LAB**.

It may be copied in full or in parts only by the Management / personnel and only for the purpose of Company related activities. Disclosure and copying of any information contained within this procedure to any person(s) outside the employee of company without written permission/authorization of the Chief Executive Officer is strictly prohibited.

 TEST METHODS AND METHOD VALIDATION MASTER CONTROLLED COPY NO. _____	Doc. No.	QCL-LSP-23
	Issue Date	10-02-2025
	Rev#	00

1.0 Purpose

This document establishes guidelines for method selection and the procedures for verification of standard / validation of non-standard method performance. Standard Testing Methods that do not need validation, but for the confidence of technical staff and commitment to quality, precision and accuracy of results laboratories verify adopted standard testing methods prior to use.

2.0 Application

This procedure covers verification of standard / validation of non-standard testing methods introduced into the Laboratories.

3.0 Terms and Definitions

- 3.1 Accuracy:** Accuracy is the nearness of a result or the mean of a set of measurements to the true value.
- 3.2 Analytical batch:** An analytical batch consists of samples which are analyzed together with the same method sequence and the same lots of reagents and with the manipulations common to each sample within the same time or in continuous sequential time periods.
- 3.3 Detection limit:** A detection limit is the lowest amount of analyte in a sample which can be detected but not necessarily quantitated as an exact value. It is often called the limit of detection (LOD) which is the lowest concentration level that can be determined statistically different from a blank at a specified level of confidence. It is determined from the analysis of sample blanks. Method detection limit (MDL) is the minimum concentration of a substance that can be measured and reported with 99% confidence that the analyte concentration is greater than zero. It is determined from analysis of a sample in each matrix containing the analyte.
- 3.4 Limit of Quantitation:** This is the level above which quantitative results may be determined with acceptable accuracy and precision.

 TEST METHODS AND METHOD VALIDATION MASTER CONTROLLED COPY NO.	Doc. No.	QCL-LSP-23
	Issue Date	10-02-2025
	Rev#	00

- 3.5 Linearity:** Linearity is the ability of the method to elicit results that are directly proportional to analyte concentration within a given range.
- 3.6 Non-standard method:** This refers to a method that is not taken from authoritative and validated sources. This includes methods from scientific journals and unpublished laboratory-developed methods.
- 3.7 Precision:** Precision is the agreement between a set of replicate measurements without assumption of knowledge of the true value. Repeatability expresses the precision under the same operating conditions over a short period of time. Intermediate precision expresses within-laboratory variations, such as different days, different analysts, and different equipment. Reproducibility expresses the precision between laboratories.
- 3.8 Range:** A range is the interval between the upper and lower concentration of analyte in sample for which it has been demonstrated that the analytical procedure has an acceptable level of accuracy, precision, and linearity.
- 3.9 Ruggedness or robustness:** Ruggedness is a measure of an analytical procedure's capacity to remain unaffected by small, but deliberate variations in method parameters and provides an indication of its reliability during normal usage.
- 3.10 Specificity:** Specificity is the ability to assess unequivocally the analyte in the presence of components which may be expected to be present.
- 3.11 Lab Standard Procedure (LSP):** This is a method that is traceable to a recognized, validated method.
- 3.12 Method Validation:** A method validation is the process of establishing the performance characteristics and limitations of a method and the identification of the influences which may change these characteristics and to what extent.
- 3.13 Verification:** Verification is the confirmation by examination and provision of objective evidence that specified requirements have been fulfilled.

 MASTER	TEST METHODS AND METHOD VALIDATION	Doc. No.	QCL-LSP-23
		Issue Date	10-02-2025
		Rev#	00

4.0 Responsibility

- 4.1 The AQCM have the responsibility to ensure implementation of method verification procedures.
- 4.2 QCM has the responsibility to implement method verification procedures in Lab.

5.0 Procedure

5.1 Method Selection

An integral part of the laboratory quality system is the use of standard testing methods. Non-standard methods could also be used after validation.

5.2 Standard Method Verification

5.2.1 Standard methods need verification to ensure that the laboratory can perform the analysis. The laboratory's quality control program as described in Quality System Manual. Verification of an analytical method is the demonstration that a laboratory is capable of replicating with an acceptable level of performance of a standard method. Verification under conditions of use is demonstrated by meeting system suitability specifications established for the method, as well as a demonstration of accuracy and precision or other method parameters for the type of method. Method performance is accomplished by using performance characteristics such as:

5.2.1.1 Precision based on analysis of multiplicities

5.2.1.2 Accuracy based on analysis of samples of known concentrations (Certified Solutions/Reference Material)

5.2.1.3 Monitoring quality control samples, usually using Range Control Charts.

5.2.2 To carry out method verification a team of technical staff members under the supervision of AQCM is formed. The team directly reports to QCM as a team leader.

5.2.3 Team selects sample(s) of known value for method verification, and checks that the laboratory has:

5.2.3.1 Suitable/required equipment.

5.2.3.2 Chemicals/standards/solutions.

5.2.3.3 Appropriate environment.

5.2.3.4 Competent personnel for performing the standard method.

5.2.4 After identifying and acquiring resources required to execute the assignment the Team carries out method verification by following the standard testing method being verified.

5.2.5 Details of analysis including accuracy, precision, any special instructions for the use of standard testing method and a statement of whether the method has been adopted for use or will require certain modifications are documented on Standard Test Method Verification Form.

5.2.6 Accuracy is determined by analyzing % Recovery as:

$$\% \text{ Recovery} = (X / K) * 100$$

Where X: Measured Value

K: Known value

5.2.7 Precision is expressed as Relative Standard Deviation (RSD) or Relative Percent Difference (RPD) of multiple samples.

RSD is calculated from standard deviation and mean recovery, when the standard deviation is derived from multiple recovery results as follows:

$$\text{RSD} = \text{CV} = (\sigma / \bar{x}) * 100$$

Where,

RSD = Relative Standard Deviation

CV = Coefficient of variation

$$\sigma = \text{SD (Standard Deviation)} = \{ \sum (x_i - \bar{x})^2 / (n - 1) \}^{1/2}$$

$\bar{x}$ = Mean

MASTER

CONTROL NO. _____

 x_i = Measured Value ($x_1, x_2, x_3, \dots, x_n$) n = No of measurements

RPD is calculated when only two sample results are available as follows:

$$RPD = \{ (|R1 - R2|) / \bar{x} \} * 100$$

Where:

 $|R1 - R2|$ = absolute difference between the determinations

- 5.2.8 In cases where the SOP /WI is modified from the standard or existing testing procedure and protocol, it is demonstrated by the Team that the modifications do not adversely affect the precision and accuracy of the data obtained.
- 5.2.9 To implement the modification, the standard or existing method is first performed. Each major modification is then validated against the original method.
- 5.2.10 Additional step to verify performance specification is 'Mean Test' for significance of difference between the two samples means to determine degree of accuracy. The sample mean of readings taken by existing method must be close to readings taken by modified method while reading taken by modified method must be close and in an acceptable range of the true/known value of the sample.
- 5.2.11 The statement on verification of standard testing methods also includes a note about acceptability and details of modifications made to the standard testing method(s) if any.

In addition to determining the accuracy and precision the factors like number of operators, number of equipment and reproducibility with respect to time is taken into consideration for determining verification of the method keeping in view the tolerance limit of the product to be tested.

5.3 Method Validation

In addition to the parameters defined in 5.2, following parameters are required:

5.3.1 Linearity

MASTER

CONTROLLED COPY
NO.

The linearity of an analytical procedure is its ability (within a given range) to obtain test results which are directly proportional to the concentration (amount) of analyte in the sample

For the establishment of linearity, minimum 5 concentrations are recommended (from LOQ till the range last or within specified range). The graph of concentration of analyte vs. response must be plotted. The *independent variable* (concentration) must be plotted against *X-axis* and *dependent variable* (i.e., response) must be plotted against *Y-axis*.

5.3.2 LOD / LOQ

The limit of detection LOD (or detection limit, DL) is the lowest possible concentration at which the method can detect (but not quantify!) the analyte within the matrix with certain degree of confidence.

LOD is calculated in the following way:

$$\text{LOD} = 3.3 * \sigma / S$$

Where,

σ = SD of the obtained results

S = Slope of the calibration curve.

The limit of quantitation LOQ (or sometimes also referred as quantification limit, QL) is the lowest possible concentration of the analyte that can be quantified by the method in a reliable way.

LOQ is calculated as follows:

$$\text{LOQ} = 10 * \sigma / S$$

Where,

σ = SD of the obtained results

TEST METHODS AND METHOD VALIDATION MASTER	Doc. No.	QCL-LSP-23
	Issue Date	10-02-2025
	Rev#	00

S = Slope of the calibration curve.

CONTROLLED COPY
NO. _____

5.3.3 Ruggedness and Robustness

The robustness/ruggedness of an analytical procedure is a measure of its capacity to remain unaffected by small, but deliberate variations in method parameters and provides an indication of its reliability during normal usage. The guidelines below shall be followed for calculations

https://fabi.research.vub.be/sites/default/files/atoms/files/robustness_tests.pdf

5.3.4 PT Participation

Participation in the same or similar PT program and qualifying with a satisfactory Z-score is mandatory.